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Advanced cooling techniques of P.V. modules: A state of art

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ABSTRACT

The efficiency of solar systems, in particular photovoltaic panels, is generally low. The output of the P.V. module is adversely affected by their surface rise in temperature. This increase is associated with the absorbed sunlight that is converted into heat, resulting in reduced power output, energy efficiency, performance and life of the panel. The use of cooling techniques can offer a potential solution to avoid excessive heating of P.V. panels and to reduce cell temperature. This paper presents details of various feasible cooling methods, including novel and advanced solutions for P.V. panels and indicates future trends of research. Different features and capability about each cooling techniques are presented, to provide better insight and valuable guidelines for researchers who intend to study, improve or optimise any type of cooling techniques of P·V. modules.

1. Introduction

In this industrial world, people live in an energy-intensive and consumer-led environment. This has contributed to the rapid downfall of fossil fuels, which is the primary basis of electricity production. It is therefore highly necessary to find sustainable sources in order to reduce our reliance on fossil fuels. Solar is the commonly used non-conventional energy available worldwide. Sun radiation is the source of all types of renewable energy. It can be converted directly or indirectly into electrical energy either by means of photovoltaic (P·V.) or thermal collectors respectively. The solar thermal system efficiencies range between 40 and 60% while P.V. has efficiencies between 10 and 20% [1,2]. Solar cells use an only visible range of wavelengths from 380 to 700 nm(nanometres) to generate electricity. Longer wavelengths of more than 700 nm do not have sufficient energy to build electron-hole pairs [3,4]. Shorter wavelengths of radiation such as X-rays do have high photon energies, but the high-energy photons could potentially damage the photovoltaic cell through ionisation processes. Outside of the visible wavelengths, the undesired radiant energies from the Sun are subsequently converted to heat, causing the solar cell temperature to increase.

The electrical power from the solar cells is increased by reducing the operating temperature [5–7]. In addition, if the lifetime of P.V. is also extended, the total amount of electricity produced will be further improved. It is widely stated that P.V. panel efficiency varies significantly depending on the material bandgap and wavelength of the sunlight. One way to improve performance is to discover novel materials to develop efficient solar systems. The electrical power of solar P.V. declines substantially whenever the solar cell temperature is high. Hardly 5–20% of the sun rays entering the surface of Solar cell is transformed into electrical power [8,9]. While the

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remaining radiation is either transmitted backwards or absorbed in the form of heat by the cell. The absorbed heat raises its temperature up to 70 °C. Several researches have been performed to cool P.V. panel in order to improve their efficiency.

This study offers an overview of the various cooling method and its key features. The study is organised into five sections. Section 1 deals with a brief background of the topic. The environmental parameters influencing the efficiency of the module are discussed in Section 2. The different approaches of cooling P.V. panels to boost their efficiency are mentioned in Section 3. Section 4 addresses the problems and challenges in the development of cooling techniques. The final part is focused on future scope and conclusions.

2. Environmental parameters affecting module efficiency

There are various environmental parameters that have an effect on the solar panel. For example, sunlight, ambient and module surface temperature, wind speed, humidity, shading, dust, installation height, etc. Among several variables, key role-players are indeed solar irradiance and temperature.

2.1. Effect of solar irradiance

The short circuit (I_{SC}) current is affected by the amount of photons absorbed by the semiconductor material and is thus related to the light intensity [10]. The conversion efficiency is therefore fairly constant in such a way that the power output is usually associated with the irradiance, but the efficiency is reduced if the cell temperature rises (Fig. 1). The open-circuit voltage (V_{OC}) varies only marginally with the light intensity [11].

2.2. Effect of ambient temperature

The V_{OC} decreases so much with the rise in temperature of the panel above 25 °C but short-circuits current, Isc, increases only marginally (Fig. 2). The temperature effect on P.V. performance is identified as the temperature coefficient. The net result is a reduction in power output with temperature rise. The percentage of temperature coefficient indicates a shift in output as it rises or falls against the normal conditions of 25° Celsius. For illustration, if the temperature coefficient for a specific panel is −0.5%, the maximum power for every 10 °C increase will be reduced by 0.5%.

The nominal operating cell temperature (NOCT) and standard test conditions (STC) are specified in Table 1. However, the panel temperature on sunny days in the equator region reaches a peak of 70 °C. Such extreme temperature leads to the local hotspot and poor currents. Hence, the panel temperature above 45 °C leads to degradation of electrical performance and panel life. Different solar cell technologies have different responses to the temperature variations, and this difference in response has been widely investigated in the literature (Table 2). Electrical efficiency of P.V. decrease with temperature is also an important field of research and literature has included many solutions. The well-known equation is:$\eta = \eta_r[1 - \beta(T_m - T_r)]$.

β value is around 0.004 K. These systems performance is based on an STC condition of 25 °C.

For this reason, it is important to remove heat from the P.V. using appropriate cooling methods in order to solve the negative impacts of cell temperature and control the working temperature within the defined value of its manufacturer.

The typical heat balance for the deployed solar P.V. is characterised as follows (Fig. 3):

From the solar module's heat balance, it is observed that the module's convective heat loss due to the air flow over the surface of the module is about 66% of the total heat loss(Table 3).

The combined effect of high irradiance and temperature results in overheating of the panel, which may result in:

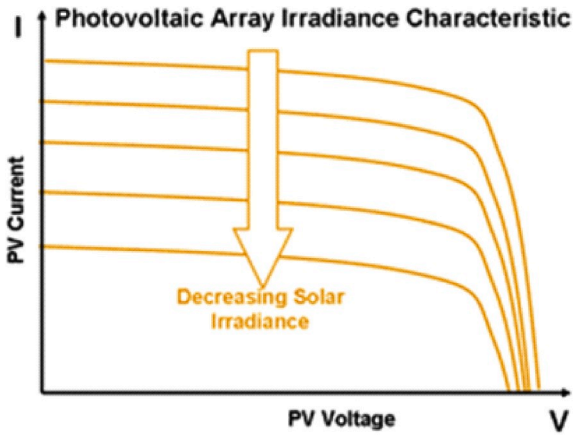


Fig. 1. Characteristics of a solar P·V.: Effect of solar irradiance.

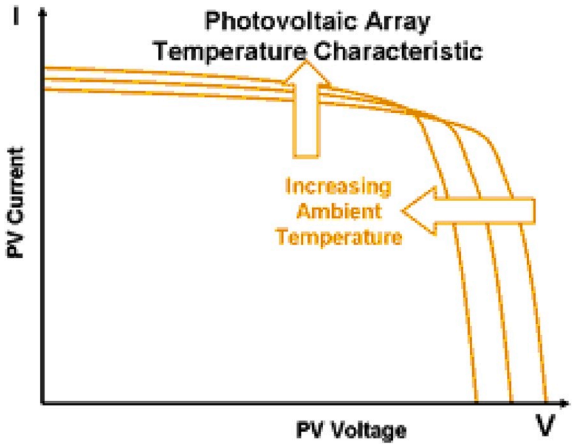


Fig. 2. Characteristics of a solar P·V.: Effect of temperature.

Table 1
STC and NOCT conditions of a Solar Photovolatic module [10].

Parameters	STC	NOCT
Solar radiation (W/m^2)	1000	800
Ambient Temperature (° C)	25	20
Air mass	1.5	–
Wind Speed	–	1
Module temperature (°C)	25	45

Table 2
Temperature coefficients, Reference efficiency of P.V. technologies [10,11].

P·V. technology	Material Thickness (°μm)	η_{ref} (%)	βref (°C^{-1})	Surface area required for 1kWp system(m^2)
Mono-cSi	**200**	16–24	0.0041	~7
Poly-cSi	**160**	14–18	0.004	~8
a-Si	**1**	4–10	0.011	~15
CIGS	**~2**	7–12	0.0048	~10
CdTe	**~1–3**	10–11	0.00035	~10

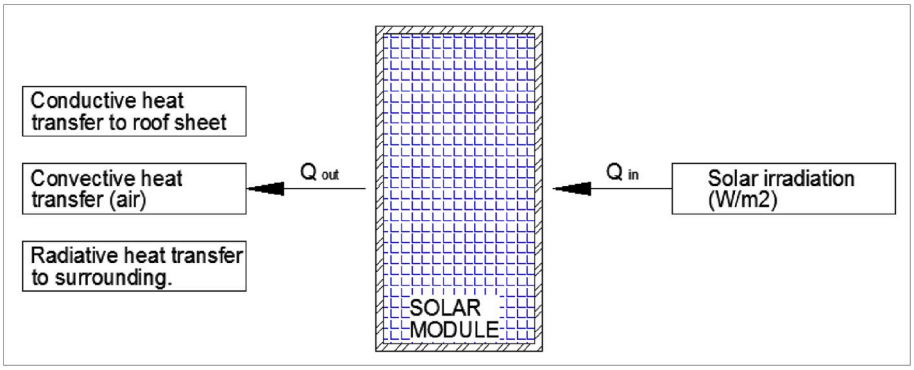


Fig. 3. Heat balance of Solar module.

- The decrease in performance.
- The decrease in conversion efficiency.
- Increase cell degradation.
- Reduce the life of the cell.

Table 3
Heat loss from P·V. module.

S.No	Heat loss from module	Percentage
1	Conduction through mounting	2
2	Convection from top surface	42
3	Convection from bottom surface	24
4	Radiation from top surface	21
5	Radiation from bottom surface	11

2.3. Effect of dust/shading

The increase in temperature of photovoltaic (P·V.) module is not only due to the climatic environment (ambient temperature) but also to the problems of direct and indirect partial shading; several recent studies are of interest to our present research [10,11]. The shading on the photovoltaic module can be caused by the projection of the shadow of an object installed far from the solar panel (example: tree or candelabrum of lighting, etc.) or caused by dust deposition, if this last is uniform throughout the photovoltaic module, it could imply a thermal runaway on the PV panel. Likewise, when partial, hot spots can be created on the P.V. module. Power loss due to partial shading and hot spots can be greater than 70% [12]. Dust is characterised as pulverised shape tiny particles of size less than 500 μm. The second environmental parameter is soiling, which depends on its distribution on the P.V. module (uniform or partial) and its size and density. The loss of efficiency depends on the exact mass and size of the dust particle deposition on the surface of the P.V. module. The deposition of the particles on the P.V. surface is very strong in the desert climate. Due to dust particles, solar output is decreased by nearly 40%.

2.4. Effect of humidity

The air humidity means the amount of water content in the air. The relative humidity is often used to indicate air humidity which is the amount of moisture in the air. According to several researchers generally, two conditions are considered to evaluate the effect of moisture. First, the effect of particles of water vapour on irradiance of sun rays and the second effect of humidity during entry in enclosed P.V. modules. As the temperature changes during the day, the relative humidity also changes accordingly. Humidity shows an inverse relationship with temperature.

2.5. Effect of wind

It may not be a correct concept to say that solar P.V. efficiency is directly affected by wind velocity. However, it has a major role to play in P.V. generation. When the wind flows, basically, the temperature of solar cell drops [13]. The wind cools the solar panels resulting in producing less vibration of the electrons so the electrons can carry more energy while moving to the upper state. Solar P.V. cooled by 1 °C are 0.05% more effective.

3. Solar P.V. Cooling techniques

3.1. Need for cooling

The change in surface temperature is influenced by external climate variables such as sunlight, wind velocity, moisture, atmospheric temperature and concentrated dust. Improvement of efficiency can be accomplished by reducing the operating temperature as it is more problematic to modify other parameters involved. Of example, in the construction of photovoltaic panels on the building facades, which are vertical and non-directional surfaces, solar radiation is an uncontrollable parameter. To make photovoltaics more efficient, by avoiding the issue of temperature rise, a variety of cooling techniques have been carried out and have been reviewed in a variety of various literature.

3.2. Classification of cooling techniques

Scientists are working on cooling systems for reducing solar cell operating temperatures, which are known as active and passive cooling systems. The appropriate cooling of the P.V. array tends to reduce the loss of output and increases the reliability of the P.V. module. Passive cooling and active methods of cooling are employed to improve performance of P·V. modules.

Active cooling requires a coolant, like air or water, which typically involves fan or pump power Whereas passive cooling requires no special power to cool P.V. cells [14,15]. In order to control and maintain the operating temperature, extensive research was carried out on the use of liquid coolant, air and other liquids, often water or glycols. Whether or not increased power output through active cooling will offset power consumption is an important economic factor.

Passive cooling techniques are classified into three main categories: passive cooling of the air, passive cooling of water and conductive cooling. Passive cooling might include extra components, like heat pipe or sink or exchanger to drive natural convection cooling [16]. The heat sinks were indeed high thermally conductive material. They are placed at the bottom of the solar panel to

maintain the overall transfer of heat from the panel to the local environment [14,16]. Passive cooling technologies are, therefore considered to be effective in reducing the temperature of P.V. cells, as they are relatively easy and cost-effective to produce.

Cooling technologies have evolved towards more complex approaches, which include heat sinks, microchannels, heat exchangers, phase - change materials (PCMs), nano-fluids, thermoelectric generators (TEGs) or combinations with other systems. The novel, sophisticated field relates to beam splitting (or spectrum filter) technology, which distinguishes the wavelengths used for P.V. cells from those used for the thermal conversion of the PVT system.

The different cooling techniques are shown in Fig. 4.

3.3. Air-based cooling

A passive method of cooling of P.V. panels and balance of systems by air is carried out naturally without any mechanical technique. Due to its simplicity, natural convection is perhaps the most popular method for P.V. Module Cooling, no additional materials are required, and the cost is relatively small. The passing of air over the P.V. panels removes the heat by convection and the air passing over the panel is more effective than the air moving under the P.V. panels. As mentioned, the most basic type of cooling is active air-cooling. Active air-cooling are systems that use fans or other means to create airflow. These types of systems can be made so that the waste heat generated from the solar panels can be used.

As such, the cooling of photovoltaic panels can be enhanced if metallic materials with fins are installed on P.V. panels back surface to ensure much more air circulation [17,18]. The photovoltaic temperature can be maintained lower than 40 °C by providing air-gap between walls and the P.V. system. Forced airflow strategies include open-air channel, metal frame, fins and ducts beneath P.V. panels [19,20]. Teo et al. use array ducts to considerably lower the temperature of solar panels and increase their efficiency between 12 and 14% [20].

Mazón-Hernández examined forced convection cooling, using fans for cooling the roof-mounted P.V. modules back-side (Fig. 5). The overall efficiency increases of 2% and the maximum cell temperature decrease of 15 °C was observed. The distance between module and roof, the mass flow rate of air and the local surrounding temperature were proven to have a greater impact on the P.V. performance. Active cooling by air is not used on P.V. systems as they usually stand in the open air. The spacing between panels is

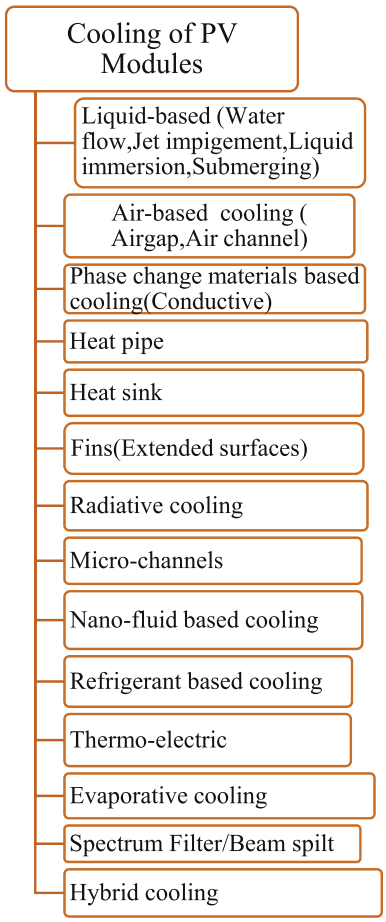


Fig. 4. Classification of cooling techniques.



Fig. 5. Picture of Natural and forced convection air cooling.

playing a vital role in the cooling of panels and balance of system due to the flow of air.

3.4. Liquid based cooling

Water is needed for the cooling system to flow continuously through the system; additional pumping energy is necessary, and the methods are bulky. Water circulation can be operated in all of these systems using solar-powered D.C. pumps in active cooling technology. Numerous technologies of water active cooling have been studied, and some effective methods are presented in this section, with their main highlights (See Table 4).

3.4.1. Forced water circulation

Heat pipes can be adopted on the rear side of module panels to minimise the temperature, even when liquid is being used as a cooling medium to cool the cells. Waste heat for P.V. modules due to excess radiation absorption can then be transferred to the circulating coolant, and even be used for other uses. According to studies (Wu 2011) [21], the overall efficiency of the system is very much dependent on pipe material heat transfer capacity. However, the implementation of such technique results in high installation and material costs and thus forced water circulation cooling process is not a suitable solution for large-scale solar plants.

3.4.2. Liquid immersion cooling

Implementing an immersion cooling technique leads to install P.V. modules underwater. High-efficiency improvement results are obtained through heat absorption by water from the P.V. panels. The performance can be improved by submerging the module in the water.

For instance, Mehrotra et al. [22] shows that with a depth of 1 cm, a 17.8% electrical efficiency enhancement can be achieved (Fig. 7). While this technique has a very little environmental impact and high-temperature reduction potential, immersion cooling is

Table 4
Advantages and Disadvantages of active water-cooling techniques.

Technique	Merits	Demerits	Temperature reduction
Water spraying	Very efficient Simple process	Partial cooling Wastage of water	Max 20 °C
Forced water circulation	Very efficient Reuse of water	High investment Equipment Degradation	20-30 °C
Liquid immersion	Highly efficient Eco-friendly	Low performance in cloudy days Ionised water impact	40 °C

not applicable to floating solar systems.

3.4.3. Water spraying

Water is sprayed through the sprinklers on the front of the P.V. modules in this system, with the pump and the connected pipes (Fig. 6). Previous research on water spraying has obtained interesting results, showing an electrical efficiency increase of up to 15% in extreme weather environments. While this system uses and wastes a large amount of water installed on the overland P.V. plant, it can be a suitable and cost-effective solution for floating solar systems.

Nizetic et al. [23] scientifically evaluated, the outcome of water spray on both sides of the P.V. panel performance. The results indicated that the front side cooling gives improved results than the back-side cooling. The electrical power improvement achieved was approximately 14.6%. A water spray technique was constructed by Moharram et al. [24] to cool solar panels. The device comprises of P.V. modules, a storage tank, a pump, spray nozzles and recycling system. With the use of water spray, the solar panel temperature reduces to 35 °C.

3.5. Phase change material (conductive)

Phase change materials (PCM) cooling is a distinct form of passive conductive cooling. PCM are substances capable of retaining thermal energy, allowing temperature stabilisation. These will absorb or release significant quantities of so-called 'latent' heat when these experiences a change in their physical state, such as during the melting and freezing cycle. PCM is listed as, organic oil, inorganic salt hydrates and eutectics. The melting temperature and latent heat are the two elements for the identification of the appropriate PCM (Fig. 8). While this cannot be considered as passive cooling in the strict sense, extra effort is not required, and heat dissipated more conductively and has the effect of retaining the same temperature. As a matter of fact, PCM is integrated with the P.V. module would become a better alternative (Fig. 9) due to its important properties for daytime heat absorption and night-time release such as.

High latent heat; High absorbing capacity before melting or solidification; The constant temperature during phase transition; Slightly higher melting temperature.

Salt-based PCMs, for example, may react with metal, causing corrosion while paraffin may react with plastics. As far as temperature is concerned, It is essential for the PCM to have a melting point above the ambient temperature at which it operates: a melting point temperature below that will trigger the PCM to liquefy before the panel is even heated, rendering it useless. Organic PCM, such as paraffin, are widely used for thermal storage systems. However, as per various studies, the structure of organic hydrocarbons has a low thermal conductivity which reduces the heat charging and discharging rate during the melting and solidification cycles. In contrast, inorganic PCMs have a higher phase change enthalpy resulting in better thermal properties. A specific PCM, CaCl2·6H2O – Fe3Cl2·6H2O eutectic, can be used for control of temperature and cooling of P·V. panels. Various recent research on PCM is highlighted in Table 5.

3.6. Heat pipe

Heat Pipe is a passive cooling device for transferring energy from the source to the sink by evaporation and condensation of the fluid in a sealed system. Typical heat pipe consisting of a sealed pipe of high thermal conductivity material, such as copper or aluminium, at both evaporator and condenser. A schematic and model of Heat pipe with solar panel is shown in Figs. 10 and 11.The heat pipe can convert heat from the solar panel to air or water, reduce the temperature and improve the efficiency of the solar panel. In certain cases, the high thermal contact resistance between both the heat pipe and the solar panel leads to lower heat transfer performance. Zhao et al. [29] developed a novel flat shape micro heat pipe array with solid solar panel contact. Tang et al. [30] have been experimentally studied the heat pipe array for P.V. cooling through air and water circulation.The temperature is reduced by 4.7 °C, and the power output rises by 8.4% for air-cooling compared to the ordinary solar panels and the temperature decreases by 8 °C and the output power increases by 13.9% for water-cooling.Summary of various research on Heat pipe is presented in Table 6.

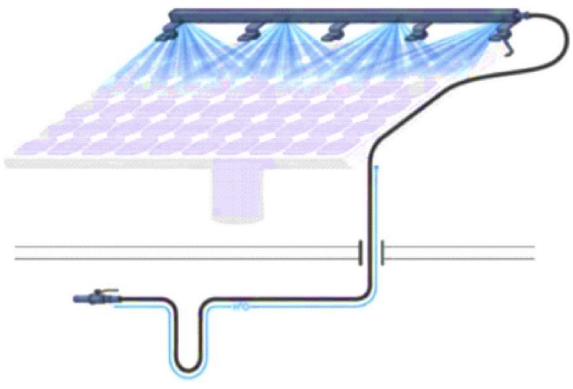


Fig. 6. A view of Water spraying [24].

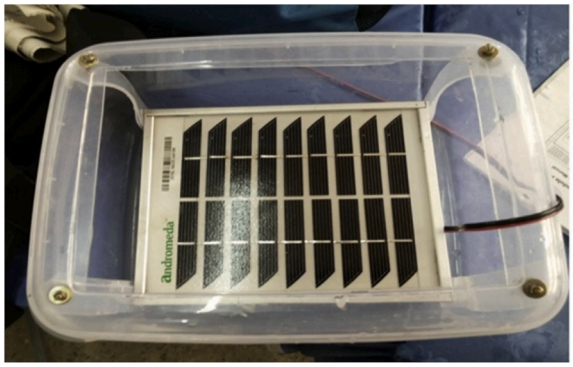


Fig. 7. A view of liquid immersion cooling [26].

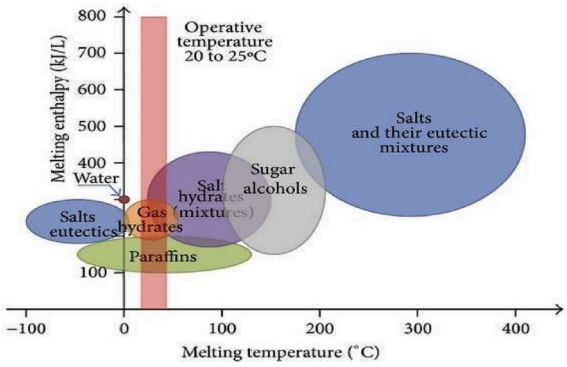


Fig. 8. Commonly used PCM and its melting temperature.

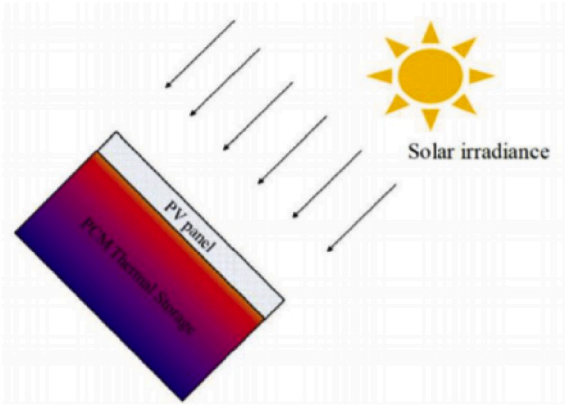


Fig. 9. Schematic of P.V. attached with PCM.

3.7. Heat sink/fins/extended surfaces/heat exchanger

Passive cooling of photovoltaic panels can be enhanced by additional components such as heat sinks, metallic materials such as fins installed on the back of P.V. to ensure convective heat transfer from air to panels [33]. The high thermal conductive heat sinks are generally located behind the solar cell. Heat sink enhances the heat transfer area from the solar cell to the ambient environment [34, 35]. It has a huge opportunity for cooling P.V. panels due to its simplicity and low cost. Very few physical tests have been carried out on the use of heat sink plates. Micro-channels are also considered to be a P.V. cooling technique used for the transfer of high heat capacity (Figs. 12 and 13). Some work on Heat sink method is listed in Table 7.

Parkunam et al. developed the strategy to improve the electrical efficiency by trying to cool the solar panel with both the support of heat sinks and a wick structure constructed of copper and aluminium fins [34].

Table 5
Summary of PCM cooling techniques.

Researcher	Type of PCM	Cell temperature	Electrical parameters	Key features
Hasan et al. [25]	PCM-RT42	Average yearly Temperature drop by 10.5 °C	Electrical output increased by 5.9%	Compare winter and summer performance
Salem et al. [26]	mixture of water and the Al2O3/ PCM	Appreciable drop in the panel temperature.	Exergy efficiency is 10.9%. Increase of 22.7%	Experimental correlations are presented.
Tan et al. [27]	Organic paraffin wax	Reduce temperature of panel by 15 °C	Improvement in Electrical efficiency by 5.39%	PCM Orientation influence the performance
Karthick et al. [28]	$CaCl_2{\cdot}6H_2O$–$Fe3Cl_2{\cdot}6H_2O$ eutectic	The maximum temperature difference was 9 °C.	Average energy gain of PV-PCM is 103.6 Whr.	ANSYS (fluent) was used to perform simulation

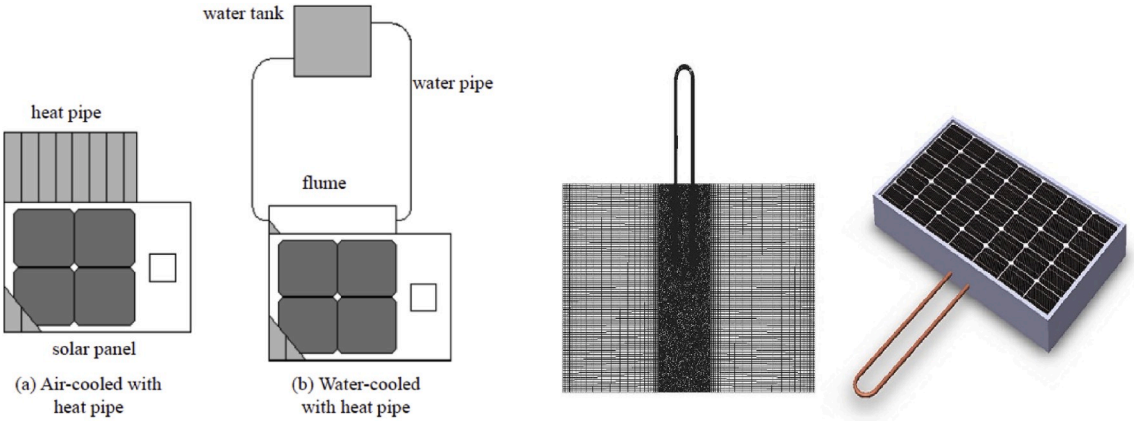


Fig. 10. Schematic of Air cooled/water-cooled heat pipe [30].

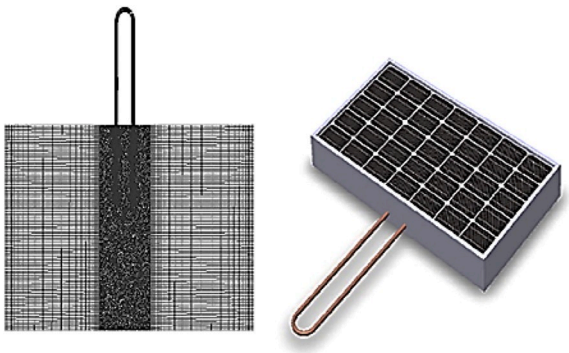


Fig. 11. Model of P.V. panel with Heat pipe.

Table 6
Summary of Heat pipe cooling techniques.

Researcher	Type of cooling system	Cell temperature	Electrical parameters	Key findings
Tang et al. [30]	Heat pipe array	4.7 °C reduction:Air cooling 8 °C: water cooling.	Power output rise by 8.4% and 13.9%	Air and Water cooling are compared.
Alizadeh et al. [31]	Single turn pulsating Heat pipe	Reduce temperature of by 16.1 K	18% enhanced power generation.	reduces the inefficiencies in a shorter time.
Habeeb [32].	Thermosyhon Heat pipe	15–35% reduction.	11–14% enhancement	MATLAB validation with experimental data.

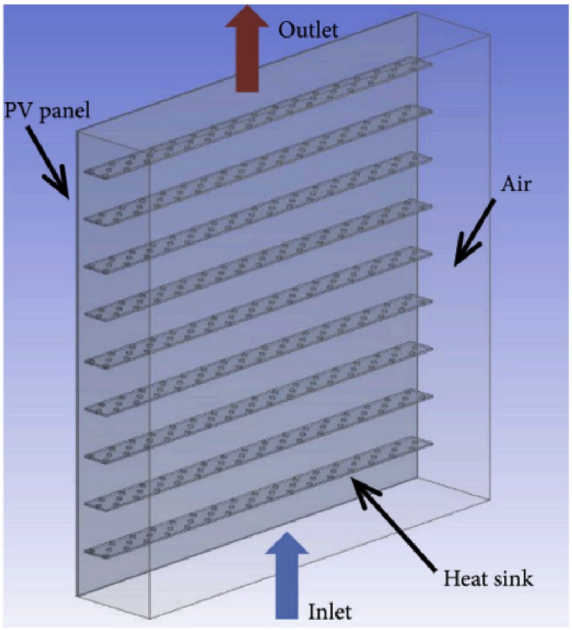


Fig. 12. Model of a P.V. panel with heat sink [36].

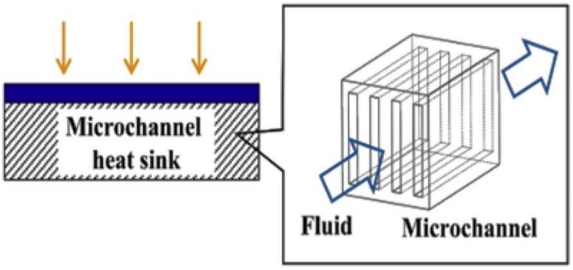


Fig. 13. Schematic diagrams of microchannel heat sink.

Table 7
Summary of Heat sink cooling techniques.

Researcher	Type of cooling system	Cooling agent	Cell temperature	Electrical parameters	Key findings
Parkunam et al. [34]	Heat sink & Wick structure	Copper and Aluminium fins	uniform module temperature for C.U. fins was obtained than aluminium fins	6.05% rise in electrical characteristics –	Efficiency and electrical performance of the copper are better than aluminium fins.
Firoozzadeh et al. [35]	Heat sink	Fins	Temperature reduction up to 7.4 °C	Leads to 2.72% increasing in efficiency. –	Both experimental and economical studies were conducted
Arifin et al. [36]	Heat sink	Air cooled	Temperature reduction from 85.3 °C to 72.8 °C.	Increased (V_{OC}) and (MPP) by 10% and 18.67%	CFD simulation was conducted on an aluminium plate heat sink with perforated fins.

3.8. Microchannel heat exchanger

Heat exchanger technique is one of a variety of common cooling methods. Heat transfer from one medium to another medium, or from one fluid to another, in heat exchanger technique. Gain maximum efficiency at high temperatures. An optimal cooling medium flow rate was performed with CFD Analysis to achieve maximum efficiency. Micro-Channel Cooling could be a feasible method to keep the P.V. Cells running within their maximum range of performance. Micro-channels can also be used as a hybrid system for PVT (Photovoltaic-Thermal Systems). Aluminium fins can be used with the aid of wind currents to cool the P.V. cells. Scaling up of the size

of the experimental set-up and manufacturing the micro-channels as efficiently as possible are the major challenge in development of Microchannel.Various research on micro-channel are summarised in Table 8.

3.9. Radiative sky cooling(photonic cooling)

The combination of atmospheric visibility in the 8-to-13-μm wavelength band with the capacity of outer space as a heat sink has provided a new paradigm in P.V. cooling. The radiative theory requires the use of the difference in temperature between outer space and earth objects. The radiative function helps to cool down when things face up to the Sun. The radiative cooling technique has also been recently studied and implemented to lower photovoltaic temperatures (Fig. 14).

Zhu et al. [40] have been tested for radiative cooling experimentally by solar absorbers. A transparent photonic structure in the wavelength of sunlight range, acts as a black body in the thermal wavelength range, was conclusively demonstrated. When a photonic structure, is placed beneath solar panel, it can radiately cool without influencing the solar absorption. The results indicated that; as a result of radiative cooling, the cell temperature dropped to 13 °C. Radiative sky cooling is a kind of a passive cooling mechanism. It is used by the thermoelectric generator-based system to preserve the cold side several degrees below normal to exploit the temperature difference between Earth and outer space [41].

3.10. Nano-fluid based cooling

Various nanoparticles mixed with water are used as nano-fluids to cool the P.V. panel and enhance system performance [42]. The various nanoparticles, are used in different weight percentages to enhance heat transfer [42–44]. Few works carried out on nano-based cooling are tabulated in Table 9.

Nano particles used are as follows:

- Boehmite,
- Aluminium oxide (Al2O3)
- Zinc oxide (ZnO)
- Titanium oxide (TiO2),
- Magnetite (Fe3O4),
- Silicone carbide (SiC), and
- copper oxide (CuO)

3.11. Thermoelectric (peltier) cooling

Thermoelectric (T.E.) cooling is yet another strategy used to decrease the P.V. panel temperature. The thermoelectric module is being used to capture and transform excess heat from the P.V. panel into electric power. The system is based on the Peltier effect, is being used to cool the P.V. panels using electrical energy. The basic principles of thermoelectric cooling are the Peltier effect concepts, as well as the see back effect. In a specific direction, the Peltier effect arises as a heat flow in an electrified junction. It causes heating at one side of the junction, and cooling impact at the other. The flow rate for heating/cooling depends on the differences in temperature and voltage/current. Until now, the PV-TEC methods are described in a limited manner.A schematic of PV cell with TE module is shown in Fig. 15.

Najafi and Woodbury [45] modelled the P.V. cell cooling with a Peltier element. The integration of thermoelectric cooling for high concentration P.V. cells has been shown to be viable.

The thermoelectric cooling of a P.V. module was experimentally studied by Borker et al. [46]. The results revealed that the performance improvement of P.V. panel due to T.E. cooling from the range 8.35–11.46% to 12.26–13.27%.

Benghanem et al. [47] observed that the temperature of the P.V. cells decreased from 83 °C to 65 °C with T.E. modules. The T.E. system delivers cooling effect with electric power derived from the panels. To be reliable, therefore, the T.E. module's energy consumption should be significantly smaller than the electricity produced by cooling P.V. panels [48].

3.12. Evaporative cooling

Evaporative cooling is a perfect choice by using water evaporation to cool the panels in the same way that sweat cools down our skin on a hot day. It works on water evaporation with the surrounding air to achieve phase equilibrium. Evaporative cooling is the most practical, useful and high-potential means to control the heat from the P·V. panels.Few studies on Evaporative cooling are given in Table 10.

Though the expected beneficial effects of evaporative cooling seem high, only a few studies have adapted this method of cooling. The significant aspects of evaporative cooling are:

- Ease of implementation.
- Much lower cost
- The strategy works well in hot and dry climates (see Fig. 16).

Table 8
Summary of Micro-channel cooling techniques.

Researcher	Type of cooling system	Cooling agent	Cell temperature	Electrical performance	Key findings
Wang et al. [37]	Microchannel Heat sink with V-ribs	Nano-fluids	Improved heat transfer performance	–	chaotic mixing theory is used for simulation.
Saadi et al. [38]	Microchannel with triangular ribs.	Air	For channel height to H = 20 mm, the P.V. temperature decreases by 6 °C	Electrical efficiency remains in a narrow range of 10–11.25%.	A blocking ratio of h/H = 0.75 seems to provide the best overall performance.
Ali et al. [39]	Micro-channelled plate	Water	Module temperature drop of 15 °C	Power improvement around 14%	Flow rate of 3 L per minute. Agreement observed between CFD and experiment

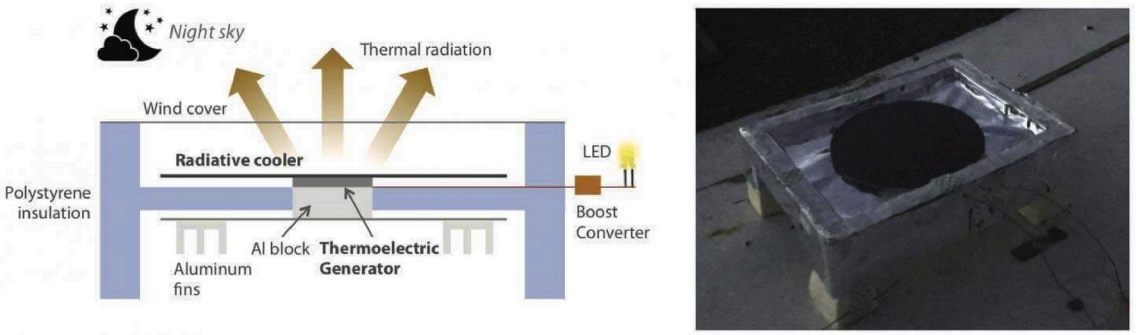


Fig. 14. Passive Radiative sky cooling(Credit: Aaswath Raman &Wei Li).

Table 9
Summary of Nano-based cooling techniques.

Researcher	Type of cooling system	Cooling agent	Cell temperature	Electrical performance	Key features
Rostami et al. [42]	nano-fluids and ultrasonics P·V. module	Atomized CuO nano-fluid	PV temperature decreases up to 57.25%	power reached to maximum of 51.1%	0.01–0.8% concentration; Flow rate from 0.4 to 12.5 m^3/h .1000 W/m2 illumination
Hussein et al. [43]	PVT system	Al_2O_3-water nano-fluid	panel temperature is reduced from 79 °C to 35 °C	50% increase in power	Concentration ratio 0.3% Mass flow rate:0.2 L/s.
Sardarabadi and Fard al [44].	PVT system	Al2O3-,TiO2,ZnO in deionised water	ZnO/Water: Outlet Temp - 3.11 °C	ZnO/Water: Electrical Efficiency 6.46%	Various metal-oxides/water nano-fluids. Experimental and Numerical studies

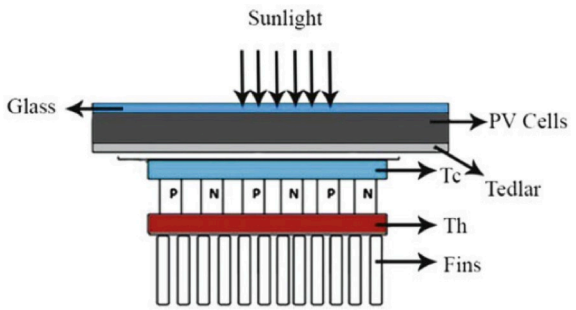


Fig. 15. Schematic of P.V. cell with T.E. module [48].

3.13. Spectrum filter cooling (optical beam spiltter)

Fluids such as vapour, organic liquids, nano-fluids can be used on Solar P.V. systems as selective absorbing fluids or as optical filters [53,54]. Optical beam split strategies are to use a liquid as the optical filter between sunlight and solar cell (Fig. 17). Liquid filters can

Table 10
Summary of Evaporative cooling.

Researcher	Type of cooling system	Cooling agent	Cell temperature	Electrical performance	Key features
Alami [49]	evaporative	synthetic clay	reference panel temperature to be high (85 °C).	15% increase in output power; 16.7% increase in output voltage	study did not show the actual variation with time of the Panels temperatures.
Suresh et al. [50]	Low cost evaporative	Cooled air	–	increase in efficiency by 10%	actual temperature variation for the module were absent
Haidar al [51].	Evaporative cooling on P·V.	Air	P.V. panel temperature by around 6 °C	–	model involved heat and mass transfer air and P·V. panel
Lucas et al. [52]	evaporative chimney	Air	temperature drop exceeded 8 °C	average electrical output increase was 4.9–7.9%.	The average ambient temperature is under 30 °C.

possibly be monitored automatically with pumps, magnetic/electric fields and temperature changes over the solar wavelengths from ultraviolet to near-infrared.

The following clear fluids were used as optical filters [55–57].

- coconut oil,
- silicone oil,
- glycerine,
- ethylene glycol etc.

Such liquid spectrum filters are installed onto a P.V. module that is thermally shielded. The filters delivers only useable spectrum to the P.V. while others are filtered out. Based on an experiment on P.V. cell with various optical filter, coconut oil performed better than water and silicone oil.

3.14. Hybrid/multi-concept cooling systems

A typical PV/T device consists of a P.V. module connected with the thermal absorber. In the PV/T system, the electrical and thermal results can not be greatly increased at the same time. Technically, solar collectors for hybrid PV/T can be developed to work at nearly 80% in cumulative capacity. But research into the various hybrid cooling systems using the following configurations, such as PV/T + PCM; PCM + nano-fluid; heat pipe + heat sink; heat sink + PCM, concurrently are rare.

Combining nano-fluid with PCM system based on PV/T is investigated and compared with conventional P.V. module [58]. With 0.01 wt percent, Karami and Rahimi used water-based Boehmite for cooling photovoltaic panels and a 27% increase in efficiency was observed [59]. The main features of PV/T integrated system are.

Operates within the range of 30–100 °C.

Design needs to balance the reduction in P.V. cell efficiency and higher performance temperatures of the thermal systems with

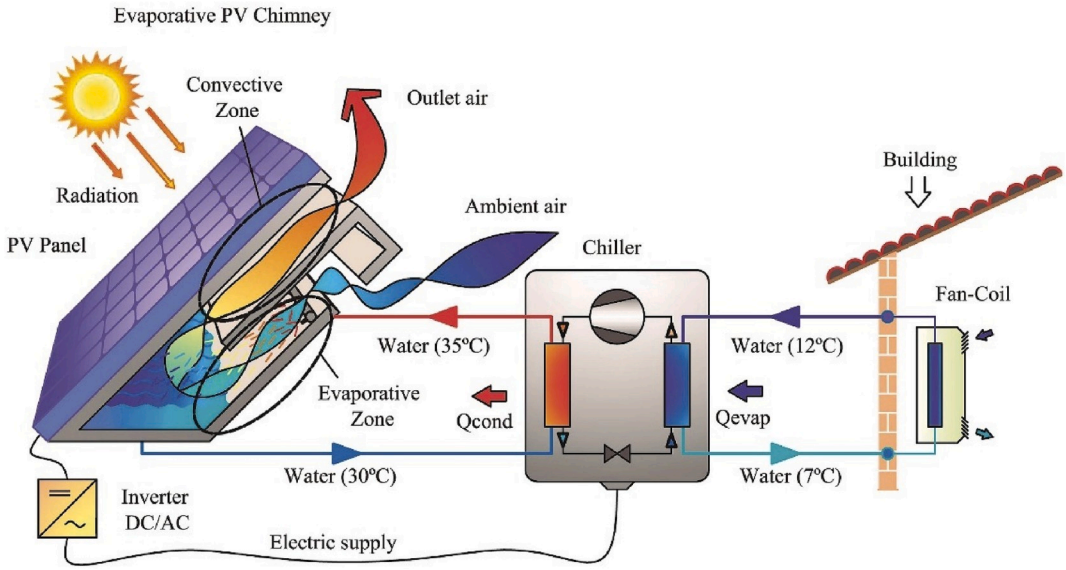


Fig. 16. Scheme of Evaporative P.V. cooling scheme proposed [52].

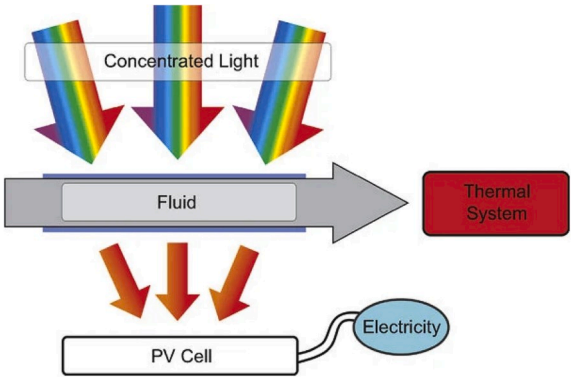


Fig. 17. Sketch of the P.V. cell using a fluid as optical filter [54].

temperature.

Technologies offer significant financial income for domestic and commercial applications with enormous electrical and thermal energy demands.

4. Challenges and future outline

Given the substantial effects of heat on Electrical efficiency of P·V., a great deal of effort was undertaken to identify cost-effective ways of cooling P.V. modules. Below is a list of the challenges of developing a cost-effective P.V. cooling system.

1. The need to recognise and understand the various variables affecting cell temperature and how potential cooling systems would be affected. It is important to design system depending on multiple factors, such as module orientation, location and cooling system components
2. The high surface area to cool with due consideration to the extremely low power output per module.
3. To ensure an even temperature distribution across the working surface, as the hotspot increases the degradation of the modules.
4. Balancing the greater initial cost with higher performance. Therefore any effective cooling system must be extremely inexpensive as it does not drive the cost of the system upward significantly.
6. If a system is designed without considering environmental influences, the device maintenance costs could outweigh the benefits of improved power output.
7. There are several parameters and variables for testing the efficiency of P.V. cooling systems, though the lack of test standards.

If a cooling system is to be successful in maximising the cooling, these challenges must be addressed while designing a P.V. cooling system. In summary, cost-effectively cooling P.V. modules present a number of challenging challenges that have led to ongoing research in this field, further improvement opportunities.

Air and water-based technologies are fairly mature, and have already been widely documented, among all cooling system. However, refrigerant-based systems and heat pipes continue to undergo, but there are still some technical/cost issues prohibiting their large scale use. As cooling P.V. panel devices, heat sinks have a high potential and should be further established. However, research regarding the evaporative cooling for P.V. panels is very limited. There is still the future for TEG in this respect, but considerable advancement is required to be accessible for commercial use. Such a system will have to focus mainly on efficiency but also on the TEG ′s costs. The progress of photonic approaches to solar cell cooling offers great opportunities, as evidenced by the above description. However, the established methods, exploit just part of the opportunities that can be made available by photonic engineering for solar cell cooling.

From a technical and economic perspective, passive air cooling with fins may be the viable option. PCM based cooling could be commercially feasible if cost reduces substantially. Passive cooling and other active cooling methods were not economically viable for the PCM. water-based active cooling system may offer an economically viable solution based on system optimisation and intelligent control of the process. A hybrid technique involving at least two of these techniques may, however, be used concurrently to lower the temperature, which would be more impressive than the approach itself.

5. Conclusion

The aim of this study was to compare the most promising PV cooling methods, with the hope to gain proper scope in design, application and future development of cooling techniques in photovoltaic systems. The following are the significant findings from the analysis of the different P.V. cooling systems.

- The findings revealed that active water cooling is the easiest and effective cooling technique and should continue to be pursued. However, active water cooling is often not practical. For worthwhile active water cooling the environment must have a steady supply of cool water, and the array to be cooled must be large to offset the energy consumption.
- The passive cooling of the P.V. system using natural convection has been observed to be the simplest way; That being said, this technique has limited possibilities. Furthermore, the air is a less effective coolant than water
- Cooling systems based on water are more ideal for application where hot water and energy requirements match like restaurants, hotels and process industries.
- Another advantage of this system is the cleaning of P.V. module against dust accumulation. In addition, it can be used in the agricultural area; the water used for cleaning can be recovered in order to be used for watering agricultural crops
- PCM cooling is the most capable technique owing to its higher energy density per unit volume,
- It can be inferred that both air and water cooling methods have been used to a large extent since they can provide additional thermal energy that can be used for different purposes.
- Although many more methods are available, the performance effectiveness of these cooling systems has to be assessed for viability. Refining, Optimising and scaling up of each method, so that large scale Solar farms can be implemented as needed.
- It is very difficult to identify a simple and cost-effective system among the proposed solutions due to lack of economic effectiveness data. Future research should focus on economic analysis and its environmental assessment.
- The future direction of technology development should focus on developing hybrid cooling methods with the main objective to maintain the surface temperature low and stable. Future research should aim to focus on one of the two promising cooling techniques, active water cooling and combined heat pipe and sink cooling.

Declaration of competing interest

The authors of the manuscript titled "**Advanced cooling techniques of PV modules: A state of Art.** Declare no conflict of interest of any of the authors with any organization in the subject matter discussed in the manuscript.

CRediT authorship contribution statement

Pushpendu Dwivedi: Conceptualization, Writing - original draft, Resources. **K. Sudhakar:** Supervision, Visualization, Writing - review & editing. **Archana Soni:** Supervision, Formal analysis, Visualization. **E Solomin:** Writing - review & editing, Visualization. **I Kirpichnikova:** Writing - review & editing, Resources, Formal analysis.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.csite.2020.100674.

References

[1] M.A. Bashir, H.M. Ali, K.P. Amber, M.W. Bashir, A. Hassan, S. Imran, M. Sajid, Performance investigation of photovoltaic modules by back surface water cooling, Therm. Sci. (2016) 1–110.
[2] M.A. Bashir, H.M. Ali, M. Ali, S. Khalil, A.M. Siddiqui, Comparison of performance measurements of photovoltaic modules during winter months in Taxila, Pakistan, Int. J. Photoenergy 2014 (2014), 898414, https://doi.org/10.1155/2014/898414.
[3] J. Siecker, A review of solar photovoltaic systems cooling technologies, Renew. Sustain. Energy Rev. 79 (2017) 192–203.
[4] F. Grubišić-Čabo, S. Nižetić, T.G. Marco, Photovoltaic Panels: a Review of the Cooling Techniques, vol. 1, 2016, pp. 63–74, 1.
[5] B. Koteswararao, K. Radha, P. Vijay, N. Raja, Experimental analysis of solar panel efficiency with different modes of cooling 8 (3) (2016) 1451–1456.
[6] M. Hasanuzzaman, Global advancement of cooling technologies for P.V. systems: a review, Sol. Energy 137 (2016) 25–45.
[7] D. Du, J. Darkwa, G. Kokogiannakis, Thermal management systems for pho- tovoltaics (P.V.) installations: a critical review, Sol. Energy 97 (2013) 238–254.
[8] L. Dorobanţu, M.O. Popescu, C.L. Popescu, A. Crăciunescu, Experimental assessment of PV panels, front water cooling strategy 1 (11) (2013) 1–4, https://doi.org/10.24084/repqj11.510.
[9] I. Ceylan, A.E. Gürel, H. Demircan, B. Aksu, Cooling of a photovoltaic module with temperature controlled solar collector, Energy Build. 72 (2014) 96–101, https://doi.org/10.1016/j.enbuild.2013.12.058.
[10] W. Luo, Y.S. Khoo, P. Hacke, V. Naumann, D. Lausch, S.P. Harvey, J.P. Singh, J. Chai, Y. Wang, A.G. Aberle, S. Ramakrishna, Potential-induced degradation in photovoltaic modules: a critical review, Energy Environ. Sci. 10 (1) (2017) 43–68, https://doi.org/10.1039/C6EE02271E.
[11] B.V. Chikate, Y. Sadawarte, The factors affecting the performance of solar cell, Int. J. Comput. Appl. Sci. Technol. (2015) 975–8887.
[12] R. Conceicao, et al., Saharan dust transport to Europe and its impact on photovoltaic performance: a case study of soiling in Portugal, Sol. Energy 160 (2018) 94–102.
[13] J.K. Kaldellis, M. Kapsali, K.a. Kavadias, Temperature and wind speed impact on the efficiency of P.V. installations. Experience obtained from outdoor measurements in Greece, Renew. Energy 66 (2014) 612–624, https://doi.org/10.1016/j.renene.2013.12.041.

[14] H.G. Teo, P.S. Lee, M.N. Hawlader, An active cooling system for photovoltaic modules, Appl. Energy 90 (2012) 309–315.
[15] S. Wu, C. Xiong, Passive cooling technology for photovoltaic panels for domestic houses, Int. J. Low Carbon Technol. 9 (2014) 118–126.
[16] M. Salih, Performance enhancement of P.V. array based on water spraying technique, International Journal of Sustainable and Green Energy (2014).
[17] Z.A. Haidar, J. Orfi, Z. Kaneesamkandi, Experimental investigation of evaporative cooling for enhancing photovoltaic panels efficiency, Results in Physics 11 (2018) 690–697.
[18] D. Sato, N. Yamada, Review of photovoltaic module cooling methods and performance evaluation of the radiative cooling method, Renew. Sustain. Energy Rev. 104 (2019) 151–166.
19 H.G. Teo, P.S. Lee, M.N.A. Hawlader, An active cooling system for photovoltaic modules, Appl. Energy 90 (1) (2012) 309–315.
[20] R. Mazón-Hernández, J.R. García-Cascales, F. Vera-García, A.S. Káiser, B. Zamora, Improving the electrical parameters of a photovoltaic panel by means of an induced or forced air stream, Int. J. Photoenergy 2013 (2013) 10. Article ID 830968.
[21] S.Y. A Wu, Q.L. Zhang, L.X. Xiao, F.H. Guo, A heat pipe photovoltaic/thermal (PV/T) hybrid system and its performance evaluation, Energy Build. 43 (2011) 3558–3567.
[22] S. Mehrotra S, P. Rawat, M. Debbarma, K. Sudhakar, Performance of a solar panel with water immersion cooling technique, Int. J. Sci. Environ. Technol. 3 (2014) 1161–1172.
[23] S. Nižetić, D. Čoko, A. Yadav, F. Grubišić-Čabo, Water spray cooling technique applied on a photovoltaic panel: the performance response, Energy Convers. Manag. 108 (2016) 287–296.
[24] K.A. S Moharrama, H. Abd-Elhadyb, M.S. Kandila, A.H. El-Sherif, Enhancing the performance of photovoltaic panels by water cooling, Ain Shams Eng. J. 4 (2013) 869–877.
[25] J. Sarwar Hasan, H. Alnoman, S. Abdelbaqi, Yearly energy performance of a photovoltaic-phase change material (PV-PCM) system in hot climate, Sol. Energy 146 (2017) 417–429, https://doi.org/10.1016/j.solener.2017.01.070.
[26] M.R. Salem, M.M. Elsayed, A.A. Abd-Elaziz, K.M. Elshazly, Performance enhancement of the photovoltaic cells using Al2O3/PCM mixture and/or water cooling-techniques, Renew. Energy 138 (2019) 876–890, https://doi.org/10.1016/j.renene.2019.02.032.
[27] Lippong Tana, Abhijit Date, Gabriel Fernandes, BaljitSingh, Sayantan Ganguly, Efficiency gains of photovoltaic system using latent heat thermal energy storage, Energy Procedia 110 (March 2017) 83–88, https://doi.org/10.1016/j.egypro.2017.03.110.
[28] A. Karthick, Ramanan Pichandi, Aritra Ghosh, B. Stalin, Ramalingam, Vigneshkumar, I. Baranilingesan, Performance enhancement of copper indium diselenide photovoltaic module using inorganic phase change material, Asia Pac. J. Chem. Eng. (2020) 1–11, https://doi.org/10.1002/apj.2480.
[29] Y. H. Zhao, et al., "A Sort of Micro Heat Pipe Array and Processing Technics," Chinese Patent: 200810225649Y..
[30] X. Tang, Z. Quan, Y. Zhao, Experimental investigation of solar panel cooling by a novel micro heat pipe array, Energy Power Eng. 2 (2010) 171–174.
[31] Hossein Alizadeh, Roghayeh Ghasempour, Mohammad Behshad Shafii, Mohammad Hossein Ahmadi, Wei-Mon Yan, Mohammad Alhuyi Nazari, Numerical simulation of P.V. cooling by using single turn pulsating heat pipe, Int. J. Heat Mass Tran. 127 (2018) 203–208, https://doi.org/10.1016/j.ijheatmasstransfer.2018.06.108. Part A.
[32] Laith Jaafer Habeeb, Dheya Ghanim Mutasher, Faez Abid Muslim Abd Ali, Cooling photovoltaic thermal solar panel by using heat pipe at baghdad climate, Int. J. Mech. Mechatron. Eng. 17 (6) (2017) 171–185.
[33] S. Sargunanathan, Performance enhancement of solar photovoltaic cells using effective cooling methods: a review, Renew. Sustain. Energy Rev. (2016).
[34] N. Parkunam, Lakshmanan Pandiyan, G. Navaneethakrishnan, S. Arul, V. Vijayan, Experimental analysis on passive cooling of flat photovoltaic panel with heat sink and wick structure, Energy Sources, Part A Recovery, Util. Environ. Eff. (2019), https://doi.org/10.1080/15567036.2019.1588429.
[35] M. Firoozzadeh, A.H. Shiravi, M. Shafiee, An experimental study on cooling the photovoltaic modules by fins to improve power generation: economic assessment.Iranian (iranica), Journal of Energy and Environment 10 (2) (2019) 80–84, https://doi.org/10.5829/ijee.2019.10.02.02.
[36] Zainal Arifin, Dominicus Danardono Dwi Prija Tjahjana, Syamsul Hadi, Rendy Adhi Rachmanto, Setyohandoko Gabriel, Bayu Sutanto, Numerical and experimental investigation of air cooling for photovoltaic panels using aluminum heat sinks, Int. J. Photoenergy (2020), 1574274, https://doi.org/10.1155/2020/1574274. Article ID.
[37] Ruijin Wang, Jiawei Wang, Weijia Yuan, Analysis and optimisation of a microchannel heat sink with V-ribs using nanofluids for micro solar cells, Micromachines 10 (9) (2019) 620, https://doi.org/10.3390/mi10090620.
[38] S. Saadi, S. Benissaad, S. Poncet, Y. Kabar, Effective cooling of photovoltaic solar cells by inserting triangular ribs: a numerical study, International Journal of Energy and Environmental Engineering 12 (7) (2018) 488–494.
[39] Muzaffar Ali, HafizM. Ali, Waqar Moazzam, M. Babar Saeed, Performance enhancement of P.V. cells through micro-channel cooling, AIMS Energy 3 (4) (2015) 699–710.
[40] L. Zhu, A. Raman, K.X. Wang, M.A. Anoma, S. Fan, Radiative cooling of solar cells, Optica 1 (2014) 32.
[41] L. Zhu, A.P. Raman, S. Fan, Radiative cooling of solar absorbers using a visibly transparent photonic crystal thermal blackbody, Proc. Natl. Acad. Sci. U.S.A. 112 (2015) 12282–12287.
[42] Z. Rostami, M. Rahimi, N. Azimi, Using high-frequency ultrasound waves and nano-fluid for increasing the efficiency and cooling performance of a P.V. module, Energy Convers. Manag. 160 (2018) 141–149.
[43] H. A. Hussien, M. Hasanuzzaman, A. H. Noman, and A. R. Abdulmunem, "Enhance photovoltaic/thermal system performance by using nano-fluid," In 3rd IET International Conference on Clean Energy and Technology (CEAT) 2014, Kuching, Malaysia, November 2014.
[44] A.M. Sardarabadi, M. Passandideh-Fard, Experimental and numerical study of metal-oxides/water nano-fluids as coolant in photovoltaic thermal systems (PVT), Sol. Energy Mater. Sol. Cell. 157 (Dec 2016) 533–542.
[45] Hamidreza Najafi, Woodbury Keith, Optimisation of a cooling system based on Peltier effect for photovoltaic cells, Sol. Energy 91 (2013) 152–160, https://doi.org/10.1016/j.solener.2013.01.026.
[46] D.S. Borkar, S.V. Prayagi, J. Gotmare, Performance evaluation of photovoltaic solar panel using thermoelectric cooling, Int. J. Eng. Res. 3 (2014) 536–539.
[47] M. Benghanem, A. Al-Mashraqi, K. Daffallah, Performance of solar cells using thermoelectric module in hot sites, Renew. Energy 89 (2016) 51–59.
[48] H. Moshfegh, M. Eslami, A. Hosseini, Thermoelectric cooling of a photovoltaic panel, in: S. Nižetić, A. Papadopoulos (Eds.), The Role of Exergy in Energy and the Environment. Green Energy and Technology, Springer, Cham, 2018.
[49] A.H. Alami, Effects of evaporative cooling on efficiency of photovoltaic modules, Energy Convers. Manag. 77 (2014) 668–679.
[50] V. Suresh, V. Naviynkumar, S. Kirubakaran, Improved power output of P.V. system by low cost evaporative cooling technology, in: G 2013 International Conference on Green Computing, Communication and Conservation of Energy, ICGCE), 2013.
[51] Z.A. Haidar, J. Orfi, H. Oztop, Z. Kaneesamkandi, Cooling of solar P.V. panels using evaporative cooling, J. Therm. Eng. 2 (2016) 928–933.
[52] M. Lucas, F. Aguilar, J. Ruiz, C. Cutillas, A. Kaiser, P. Vicente, Photovoltaic Evaporative Chimney as a new alternative to enhance solar cooling, Renew. Energy 111 (2017) 26–37.
[53] X. Han, D. Xue, J. Zheng, S. Alelyani, X. Chen, Spectral characterisation of spectrally selective liquid absorption filters and exploring their effects on concentrator solar cells, Renew. Energy 131 (2019), 938e45, https://doi.org/10.1016/j.renene.2018.07.125.
[54] R. Taylor, T. Otanicar, G. Rosengarten, Nanofluid-based optical filter optimisation for PV/T systems, Light Sci. Appl. 1 (2012) e34, https://doi.org/10.1038/lsa.2012.34.
[55] SandeepS. Joshi, Ashwinkumar S. Dhoble, Experimental Investigation of solar photovoltaic thermal system using water, coconut oil and silicone oil as spectrum filters, J. Braz. Soc. Mech. Sci. Eng. 39 (8) (August 2017) 3227–3236, https://doi.org/10.1007/s40430-017-0802-0. Springer.
[56] SandeepS. Joshi, Ashwinkumar S. Dhoble, et al., Performance analysis of photovoltaic thermal system using silicone oil spectrum filter, Appl. Sol. Energy -Springer 54 (Issue-01) (2018) 4–9, https://doi.org/10.3103/S0003701X18010061.

[57] SandeepS. Joshi, Ashwinkumar S. Dhoble, Analytical approach for performance estimation of BSPVT system with liquid spectrum filters, Energy 157 (2018) 778–791, https://doi.org/10.1016/j.energy.2018.05.204.
[58] Ali Hassan, Abdul Wahab, Muhammad Arslan Qasim, Muhammad Mansoor Janjua, Muhammad Aon Ali, Hafiz Muhammad Ali, Tufail Rehman Jadoon, Ejaz Ali, Ahsan Raza, Noshairwan Javaid, Thermal management and uniform temperature regulation of photovoltaic modules using hybrid phase change materials-nanofluids system, Renew. Energy 145 (2020) 282–293, https://doi.org/10.1016/j.renene.2019.05.130.
[59] N. Karami, M. Rahimi, Heat transfer enhancement in a hybrid microchannel-photovoltaic cell using Boehmite nano-fluid, Int. Commun. Heat Mass Tran. 55 (2014) 45–52.